

Security vulnerabilities in base protocols (NDP)

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Background

Overview

- Securing base protocols
 - Implement a hardware or software solution that mitigates or eliminates a vulnerability in a base protocol such as TCP, UDP, DNS, etc.
- We chose Neighbor Discovery Protocol (NDP)
 - IPv6
 - Developed in 2007
 - RFC 4861 - <https://tools.ietf.org/html/rfc4861>

Neighbor Discovery Protocol

- NDP is a link layer protocol for IPv6 and is used for gathering routing information on the local network
 - IPv6 version of ARP (Address Resolution Protocol)
- NDP shares many of the same vulnerabilities as ARP
 - Including Spoofing, DoS, Cache poisoning
- NDP defines 5 ICMPv6 message types
 - Router Solicitation (RS)
 - Router Advertisement (RA)
 - Neighbor Solicitation (NS)
 - Neighbor Advertisement (NA)
 - Redirect

NDP Spoofing

- Like ARP, there is not any authentication with address resolution
- Malicious actor crafts RA or NA messages and sends them to a victim or broadcasts it to the network
- Detectable since mappings are not supposed to change often and advertisements are generally not spammed
 - Some spoofing attacks spam crafted messages

Implementation

Approach

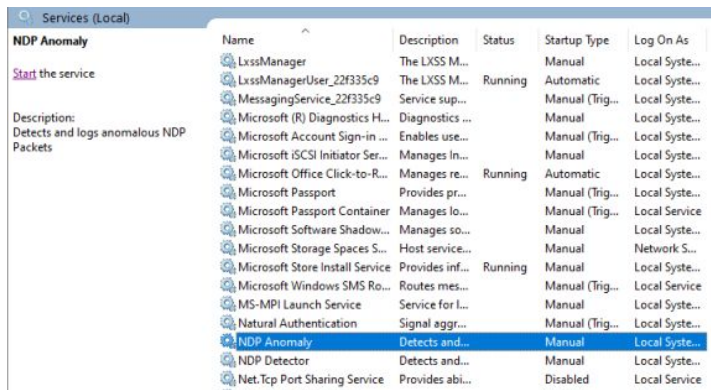
- Spoofing detection
 - Monitor local network for spoofed messages
 - Generate alerts or log messages that allows for ingestion by SIEM
- Secure Neighbor Discovery
 - Defined by IETF in RFC 3971
 - Fixes the spoofing issues by using cryptographically generated address and certificates
 - Cryptographically generated IPv6 addresses tie address to a public key

Detection

- Check source against a list of known sources
- Need a cache to store the initial list of sources and update depending on specified TTL
- Attempt to detect changes to the mappings or check if advertisements are being sent at abnormal rate

Monitoring Tool

- Code is written in Python 3.8 with the latest version of Scapy
- Can be run as a service on both Windows and Linux (systemd)



```
Apr 28 15:48:50 rbl systemd[1]: Started NDP Anomaly Detection Service.
Apr 28 15:48:50 rbl python[3534]: Hello from the NDP monitor
Apr 28 15:48:50 rbl python[3534]: Current mappings at initialization {'70:85:c2:ad:09:b5': 'fe80::a89a:bf96:5a75:6a75'}
Apr 28 15:48:54 rbl python[3534]: Packet 0 - ANOMALY DETECTED
Apr 28 15:48:54 rbl python[3534]: 70:85:c2:ad:09:b5 is trying to use fe80::a89a:bf96:5a75:6969 as IPv6 instead of previously recorded fe80::a89a:bf96:5a75:6a75
Apr 28 15:48:54 rbl python[3534]: Current mappings {'70:85:c2:ad:09:b5': 'fe80::a89a:bf96:5a75:6a75'}
Apr 28 15:48:56 rbl python[3534]: Packet 2 - ANOMALY DETECTED
Apr 28 15:48:56 rbl python[3534]: 70:85:c2:ad:09:b5 is trying to use fe80::a89a:bf96:5a75:6969 as IPv6 instead of previously recorded fe80::a89a:bf96:5a75:6a75
Apr 28 15:48:56 rbl python[3534]: Current mappings {'70:85:c2:ad:09:b5': 'fe80::a89a:bf96:5a75:6a75', '20:79:18:d3:76:06': 'fe80::7240:34ab:80ba:c64a'}
Apr 28 15:48:57 rbl python[3534]: Packet 4 - ANOMALY DETECTED
Apr 28 15:48:57 rbl python[3534]: 70:85:c2:ad:09:b5 is trying to use fe80::a89a:bf96:5a75:6969 as IPv6 instead of previously recorded fe80::a89a:bf96:5a75:6a75
Apr 28 15:48:57 rbl python[3534]: Current mappings {'70:85:c2:ad:09:b5': 'fe80::a89a:bf96:5a75:6a75', '20:79:18:d3:76:06': 'fe80::7240:34ab:80ba:c64a'}
```

Cryptographically Generated Addresses

- Sender sends advertisement with CGA along with public key
- Receiver regenerates CGA to verify that host is valid
- Requires
 - 64-bit IPv6 prefix
 - RSA key (modulus,exponent)
 - Security bit
 - (optional) Support for additional parameters

CGA Generator

- Wrote the command line utility in python
- Takes in a public key and combines that with 64bit prefix using some cryptographic methods
 - The resulting 128 bits is the address to be used
- Verification goes in the opposite direction

Future Work

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- Implement the rest of Secure NDP
- Add spamming detection to the monitoring tool
- Add additional protocols to the monitoring tool